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310405

Roll No. _____ Total No of Pages: 2

310405

B. Tech. III - Sem. (Main) Exam., February - 2021

Computer Science & Engineering

3CS4-05 Data Structures and Algorithms

Common for CS/ IT

Time: 3 Hours **Maximum Marks: 120**

Instructions to Candidates:

Part – A: *Short answer questions (up to 25 words) 10 × 2 marks = 20 marks. All ten questions are compulsory.*

Part – B: *Analytical/Problem Solving questions (up to 100 words) 5 × 8 marks = 40 marks. Candidates have to answer five questions out of seven.*

Part – C: *Descriptive/Analytical/Problem Solving questions 4 × 15 marks = 60 marks. Candidates have to answer four questions out of five.*

Schematic diagrams must be shown wherever necessary. Any data you feel missing may suitably be assumed and stated clearly. Units of quantities used/calculated must be stated clearly.

Use of following supporting materials is permitted during examination. (Mentioned in form No. 205)

1. NIL 2. NIL

PART - A

Q.1 How we can implement Queue using Array?

Q.2 Explain Basic Stack Operation.

Q.3 What do you mean by sequential search?

Q.4 Write one advantage and disadvantage of Single linked list.

Q.5 Discuss Hash function.

Q.6 Explain Tower of Hanoi Problem.

Q.7 Explain linked list with help of example.

Q.8 What do you mean by BFS?

Q.9 Discuss binary search for five elements.

Q.10 Explain Binary tree.

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PART – B

Q.1 Write and explain a program for AVL tree.

Q.2 Demonstrate the concept of Circular Queue with an example.

Q.3 Discuss a program to evaluate a given arithmetic expression: 100 * (2+12)/14?

Q.4 Explain Dijkstra's shortest path algorithm with suitable data.

Q.5 Differentiate between B tree and B+ tree.

Q.6 Write and explain program for priority queue.

Q.7 Discuss reversing of list as a stack application with an example.

PART – C

Q.1 Write a note on Double linked list. Explain operation on Double linked with the help of program and suitable data.

Q.2 Discuss step by step procedure for the following –

(a) Merge sort

(b) Selection sort

Q.3 Explain the concept of Binary tree traversal. Write code for inorder, perorder and postorder traversal of Binary tree. Give example.

Q.4 Write program of Heap sorting. Discuss step by step procedure for heap sorting for following set of data -

28 67 56 36 18 85 71

Q.5 Write and explain program to convert infix to postfix expression. Convert the following infix expression to postfix expression -

a + b * (c ^ d - e) ^ (f + g * h) – i

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